Name:____

Assume the following :

Assume I want each thread to execute the following critical section, where counter is a variable shared between all threads and is volatile (the compiler will not optimize this loop).

```
for (int i = 0; i < 100; i++)
{
    counter = counter + 1;
}
```

1. Assume you add a lock around the current critical section.
   (a) How would this affect scalability?
   (b) If you periodically check the value of counter throughout the for loop, would it be correct?

2. Assume each thread increments a local value and then adds that to the global value one time at the end.
   - How would this affect scalability?
   - If you periodically check the value of counter throughout the for loop, would it be correct?

3. Assume you add a sloppiness value S. Each thread updates its local value every iteration. Each S iterations, each thread increments the global counter by its local value.
   - How does the value of S affect scalability?
   - How does the value of S affect correctness throughout the program (e.g. if you check for correctness throughout the for loop).